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Distributed Systems: A Practical Primer

What every developer should know before building one.

The Eight Fallacies

01 The network is reliable

02 Latency is zero

03 Bandwidth is infinite

04 The network is secure

05 Topology doesn't change

06 There is one administrator

07 Transport cost is zero

08 The network is homogeneous

Every distributed system bug traces back to one of these.

CAP Theorem (Simplified)

Pick two:

- **Consistency** — every read returns the most recent write
- **Availability** — every request receives a response
- **Partition tolerance** — the system works despite network splits

In practice, partitions happen. So you're choosing between C and A.

Patterns That Work

- **Idempotency** — safe to retry any operation
- **Circuit breakers** — fail fast instead of waiting
- **Event sourcing** — append-only log of state changes
- **CQRS** — separate read and write models
- **Saga pattern** — distributed transactions without 2PC

The One Rule

If you can solve it without distribution, do that instead.

Distribution adds latency, complexity, and failure modes. Only distribute when you must — for scale, availability, or data locality.